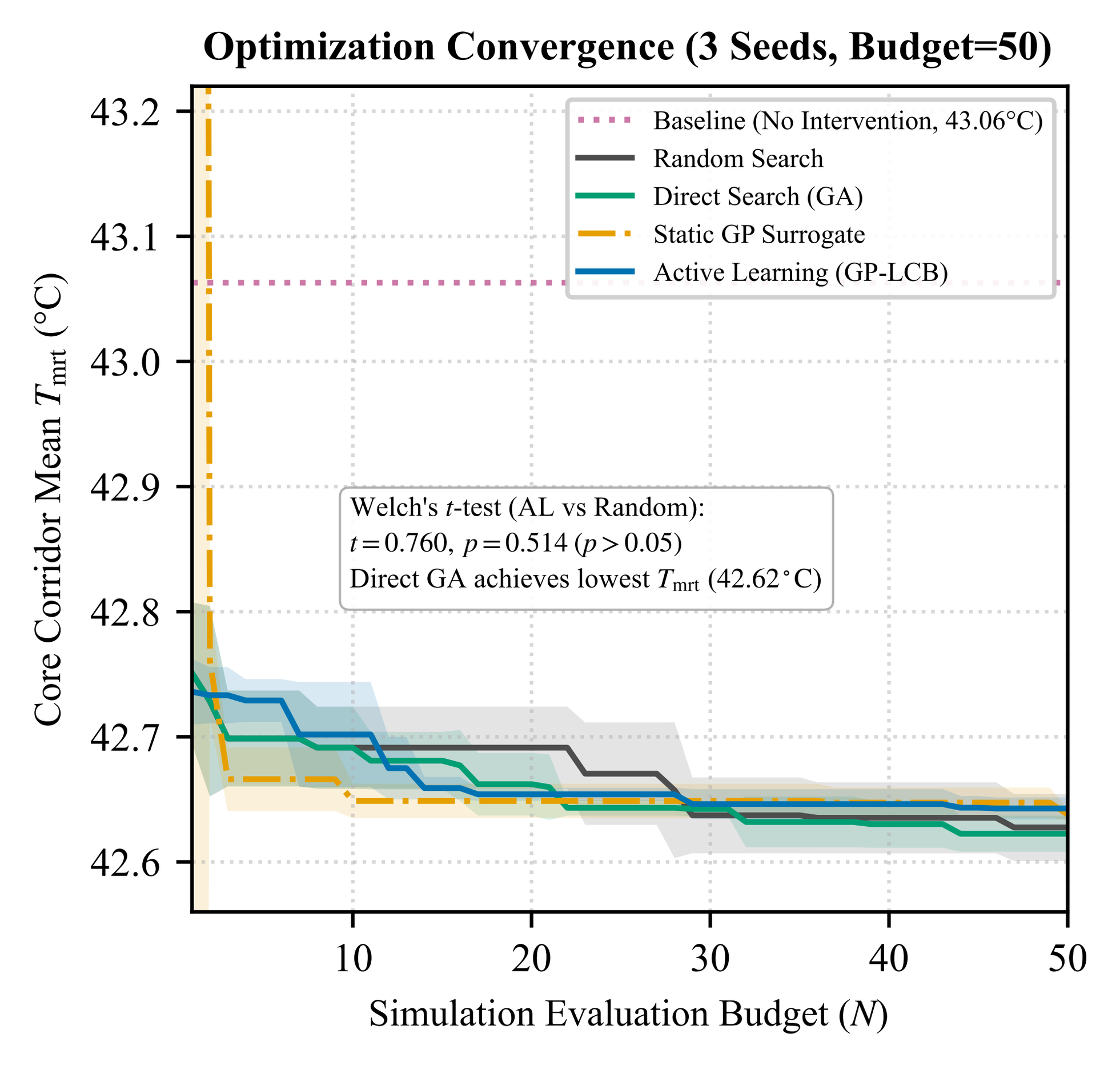
Optimization Convergence (3 Seeds, Budget=50)
Baseline (No Intervention, 43.06°C)
Random Search
Direct Search (GA)
Static GP Surrogate
Active Learning (GP-LCB)
Welch's t-test (AL vs Random):
t = 0.760, p = 0.514 (p > 0.05)
Direct GA achieves lowest T_mrt (42.62 °C)
Core Corridor Mean T_mrt (°C)
Simulation Evaluation Budget (N)
43.2
43.1
43.0
42.9
42.8
42.7
42.6
10
20
30
40
50